

TIA–FPGA – FPGA replacement for the Atari 2600 Television Interface Adaptor (C010444)

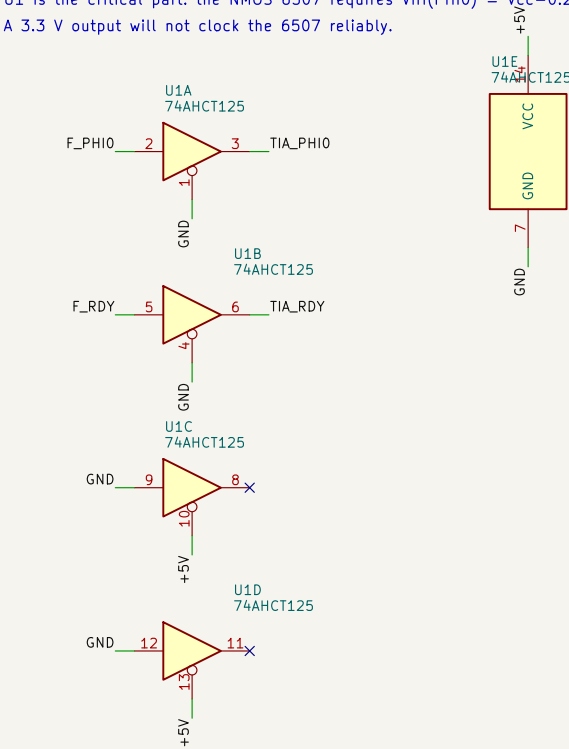
Rev A – digital interface only. Chroma and paddles do not fit the pin budget; see the notes.

FPGA MODULE – Sipeed Tang Nano 9K (GW1NR–LV9QN88PC6/I5)

Pin map verified against both Sipeed board revisions (3672 and 3674), which agree. Of the 48 header pins only 29 are usable 3.3 V GPIO: 8 are BANK3 at 1.8 V, 8 are HDMI differential pairs, 3 are power. Rev A uses all 29, with none spare.

LEVEL TRANSLATORS

U1 is the critical part: the NMOS 6507 requires $V_{IH}(PHIO) = V_{cc}-0.2\text{ V} \sim 4.8\text{ V}$. A 3.3 V output will not clock the 6507 reliably.



COLOUR VIDEO – via the Tang Nano’s HDMI connector

The 7-bit colour value never leaves the FPGA: an internal TMDS encoder drives the HDMI pins, which are dedicated differential pairs on the module and were never part of the 29-pin GPIO budget. Colour costs zero header pins. FPGA pins (P/N): CK 69/68, D0 71/70, D1 73/72, D2 75/74. RULE: leave J1/23–24 and J4/12–17 open – they carry the same TMDS pairs, and any stub degrades the signal.

TIA SOCKET – DIP–40

Two 20-way SIP strips that plug into the 2600 socket or a breadboard, in place of the C010444.

